BrightLight Tutorials _ Data Analytics

SQL Fundamentals

Exercise 3: SQL CASE Statement

Questions

1. Table: Products

Product_id	Product_name	Price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

```
Select
        Product_name,
        Price,
CASE
        WHEN price > 1000 THEN '01 Expensive'
        WHEN Price between 100 and 1000 THEN '02 Mid-range'
        WHEN Price < 100 THEN '03 Budget'
        ELSE '04 No Budget'
        END AS PRICE_Category
From products;
```

__ Return Results

Product_name	Price	Price_Category
Laptop	1200.00	'Expensive'
Phone	800.00	'Mid-range'
Keyboard	45.00	'Budget'
Monitor	300.00	'Mid-range'
Mouse	25.00	'Budget'

2. Table: Orders

order_id	Customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

```
Select
        Customer_name,
        amount,
CASE
        WHEN amount ≥ 1000 THEN 'High Value'
        WHEN amount between 500 and 999.99 THEN 'Medium Value'
        WHEN amount < 500 THEN 'Low Value'
END AS Order_Value_Category
From Orders;
```

-- Return Results:

customer_name	amount	order_value_Category
Alice	150-00	'Low Value'
Bob	560-00	'Medium Value'
Charlie	999.99	'Medium Value'
Diana	45.50	'Low Value'
Ethan	1200-00	'High Value'

3. Table: employees

emp_id	emp_name	department	Salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

```
Select
        emp_name,
        department,
        Salary,
CASE
        WHEN department = 'IT' and Salary > 80000 THEN 'Senior IT'
        WHEN department = 'HR' and Salary > 55000 THEN 'Experienced HR'
        ELSE Staff
END AS position_level;
From employees;
```

— Return Results

emp_name	department	Salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Table: Students

student_id	Student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

```
Selec
        Student_name,
        Score,
CASE
    WHEN Score ≥ 90 THEN 'A'
    WHEN Score between 80 and 89 THEN 'B'
    WHEN Score between 70 and 79 THEN 'C'
    WHEN Score between 60 and 69 THEN 'D'
    WHEN Score < 60 THEN 'F'
END AS grade;
From students;
```

— Return Results

student_name	score	grade
Ana	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. Table: deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

```
Select
        delivery_id,
        delivery_time_minutes,
CASE
        WHEN delivery_time_minutes ≤ 30 mins THEN 'Fast'
        WHEN delivery_time_minutes between 31 and 60 mins THEN 'On Time'
        WHEN delivery_time_minutes > 60 mins THEN 'Late'
END AS performance
From deliveries;
```

... return Results

delivery_id	delivery_time_Minutes	Performance
1	45	On Time
2	80	late
3	30	Fast
4	65	late
5	100	late

6. Table: tickets

ticket_id	issue_type	Priority
1	login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

```
Select
        issue_type,
        Priority,
CASE
    WHEN priority = 3 THEN 'HIGH'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'
END AS Priority_label
From tickets;

-- Return Results:
```

issue_type	Priority	Priority_label
login issue	1	Low
Server down	3	High
Slow System	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Table: attendance

Student_id	days_present	total_days	attendance_percentage
1	45	50	90%
2	30	50	60%
3	48	50	96%
4	25	50	50%
5	50	50	100%

```
Select
        student_id,
        attendance_percentage,
CASE
    WHEN attendance_percentage ≥ 90% THEN 'Excellent'
    WHEN attendance_percentage between 75% and 89% THEN 'Good'
    WHEN attendance_percentage < 75% THEN 'Needs Improvement'
END AS attendance_status
From attendance;
```

Return Results:

student_id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs Improvement
3	96%	Excellent
4	50%	Needs Improvement
5	100%	Excellent

8. Table: Products_Inventory

Product_id	Stock_qty
1	5
2	0
3	25
4	10
5	3

```
Select
        Product_id,
        Stock_qty,
CASE
    WHEN Stock_qty = 0 THEN 'out of Stock'
    WHEN Stock_qty between 1 and 5 THEN 'Low Stock'
    WHEN Stock_qty = 5 THEN 'in stock'
END AS Stock_status
From products_Inventory;
```

-- Return Results

Product_id	Stock_qty	Stock_status
1	5	In Stock
2	0	out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

9. Table: Classes

Class_id	Subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

```
Select
        Subject,
        enrolled_Students,
CASE
    WHEN enrolled_students ≥ 25 THEN 'Large'
    WHEN enrolled_students between 10 and 24 THEN 'Medium'
    WHEN enrolled_students < 10 THEN 'Small'
END AS Class_Size_Category
From Classes;
```

-- Return Results

Subject	enrolled_students	Class_Size_Category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Table: Payments

Payment_id	amount	Payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash

```
Select
        Payment_id,
        Payment_Method,
        amount,
CASE
    WHEN Payment_Method = 'Cash' and amount ≥ 200 THEN 'Eligible for discount'

    Else 'Not Eligible'
END AS discount_eligibility
From payments;

-- Return Results
```

Payment_id	Payment_method	amount	discount_eligibility
1	Card	50-00	Not Eligible
2	Cash	200-00	Eligible for discount
3	Card	150-00	Not Eligible
4	PayPal	75-00	Not Eligible
5	Cash	300-00	Eligible for discount